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riya.soni17oct@gmail.com

NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » Introduction To Machine Learning - IITKGP (course)

Week 6 : Assignment 6

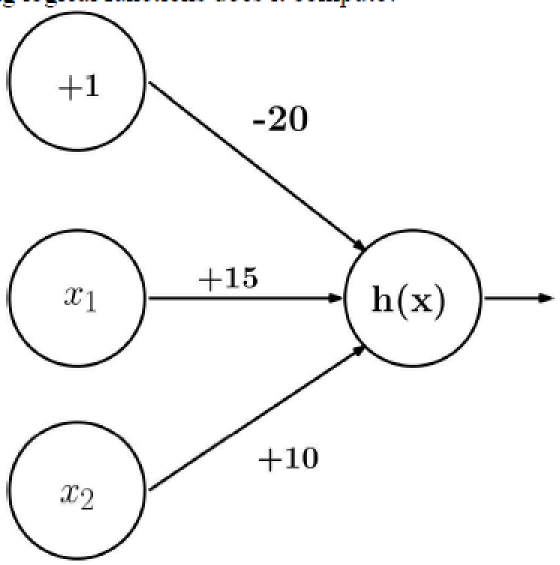
The due date for submitting this assignment has passed.

Due on 2024-09-04, 23:59 IST.

As per our records you have not submitted this assignment.

1) ***2 points***

The neural network given below takes two binary valued inputs $x_1, x_2 \in \{0,1\}$ and the activation function is the binary threshold function ($h(x) = 1\ if\ x > 0; 0\ otherwise$). Which of the following logical functions does it compute?



A) AND
B) OR
C) NAND
D) None of the above

- A)
- B)
- C)

D)

No, the answer is incorrect.
Score: 0

Accepted Answers:
A)

2) **_2 points_**

What is the sequence of the following tasks in a perceptron?

I) Initialize the weights of the perceptron randomly.
II) Go to the next batch of data set.
III) If the prediction does not match the output, change the weigh
IV) For a sample input, compute an output.

A) I, II, III, IV
B) IV, III, II, I
C) III, I, II, IV
D) I, IV, III, II

A)
B)
C)
D)

No, the answer is incorrect.
Score: 0

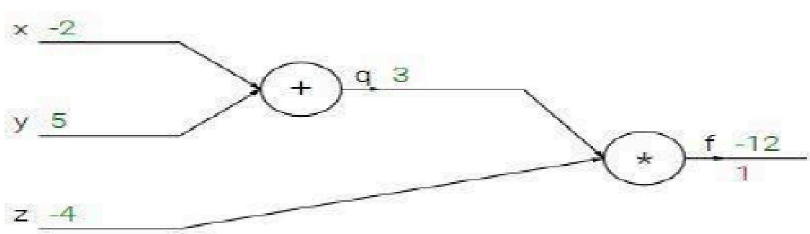
Accepted Answers:
D)

3) **_2 points_**

Suppose you have inputs as x, y, and z with values -2, 5, and -4 respectively. You have a neuron ‘q’ and neuron ‘f’ with functions:

$q = x + y$
$f = q * z$

Graphical representation of the functions is as follows:



What is the gradient of f with respect to x, y, and z?

A) (-3, 4, 4)
B) (4, 4, 3)
C) (-4, -4, 3)
D) (3, -4, -4)

A)

B)

C)

D)

No, the answer is incorrect.
Score: 0

Accepted Answers:
C)

4) **2 points**

For a fully-connected neural network with one hidden layer, what effect should increasing the number of hidden units have on bias and variance?

A. Decrease bias, increase variance
B. Increase bias, increase variance
C. Increase bias, decrease variance
D. No effect

A)

B)

C)

D)

No, the answer is incorrect.
Score: 0

Accepted Answers:
A)

5) **2 points**

Which of the following is true about model capacity (where model capacity means the ability of a neural network to approximate complex functions)?

A) As number of hidden layers increase, model capacity increases
B) As dropout ratio increases, model capacity increases
C) As learning rate increases, model capacity increases
D) None of these.

A)

B)

C)

D)

No, the answer is incorrect.
Score: 0

Accepted Answers:
A)

6) **2 points**

The back-propagation learning algorithm applied to a two layer neural network

A) always finds the globally optimal solution.
B) finds a locally optimal solution which may be globally optimal.
C) never finds the globally optimal solution.
D) finds a locally optimal solution which is never globally optimal

A)

B)

C)

D)

No, the answer is incorrect.
Score: 0

Accepted Answers:
B)

7) **2 points**

Which of the following gives non-linearity to a neural network

A) Gradient descent
B) Bias
C) Sigmoid Activation Function
D) None

A)

B)

C)

D)

No, the answer is incorrect.
Score: 0

Accepted Answers:
C)

8) **2 points**

The network that involves backward links from outputs to the inputs and hidden layers is called as

A) Self-organizing Maps
B) Perceptron
C) Recurrent Neural Networks
D) Multi-Layered Perceptron

A)

B)

C)

D)

No, the answer is incorrect.
Score: 0

Accepted Answers:
C)

9) **2 points**

A Convolutional Neural Network(CNN) is a Deep Neural Network which can extract various abstract features from an input required for a given task. Given are the operations performed by a CNN on an input:

1) Max Pooling
2) Convolution Operation
3) Flatten
4) Forward propagation by Fully Connected Network

Identify the correct sequence of operations performed from the options below:

A) 4,3,2,1
B) 2,1,3,4
C) 3,1,2,4
D) 4,2,1,3

A)
B)
C)
D)

No, the answer is incorrect.
Score: 0

Accepted Answers:
B)

10) **2 points**

In training a neural network, we notice that the loss does not increase in the first few starting epochs: What is the reason for this?

A) The learning Rate is low.
B) The Regularization Parameter is High.
C) Stuck at the Local Minima.
D) All of the above could be the reason.

A)
B)
C)
D)

No, the answer is incorrect.
Score: 0

Accepted Answers:
D)